

Three Unifying Principles

Prediction Error

Content-Addressable Memory

Active Inference

WP1: Memory & CL

Distributed memory modules

Surprise-gated write

Hierarchical time scales

Online single-pass learning

100% CORe50 · 98.2% online CL
70.3% tPC (no pretraining) · FGT=0

CORe50, Split-CIFAR, Path-X

PAM

Temporal Predictive Coding

HDC Kanerva Memory

Foundation

Extension

Design rules

Compositions

Stress test

WP2: Reasoning

Algebraic HDC composition

FOMAML meta-learning

Active inference

Causal world models

24.7% ARC ($2 \times$ GPT-4o)
100% analogies · 99.5% CLUTRR

ARC, letter analogies, Ravens, bAbI

WP3: Long-Horizon Agents

Memory Maze ($T=10,000$) · Minecraft · Factorio

3-level policy hierarchy · Adaptive curriculum

Causal interventions · Module specialisation analysis